

Machine Learning Model Performance Report

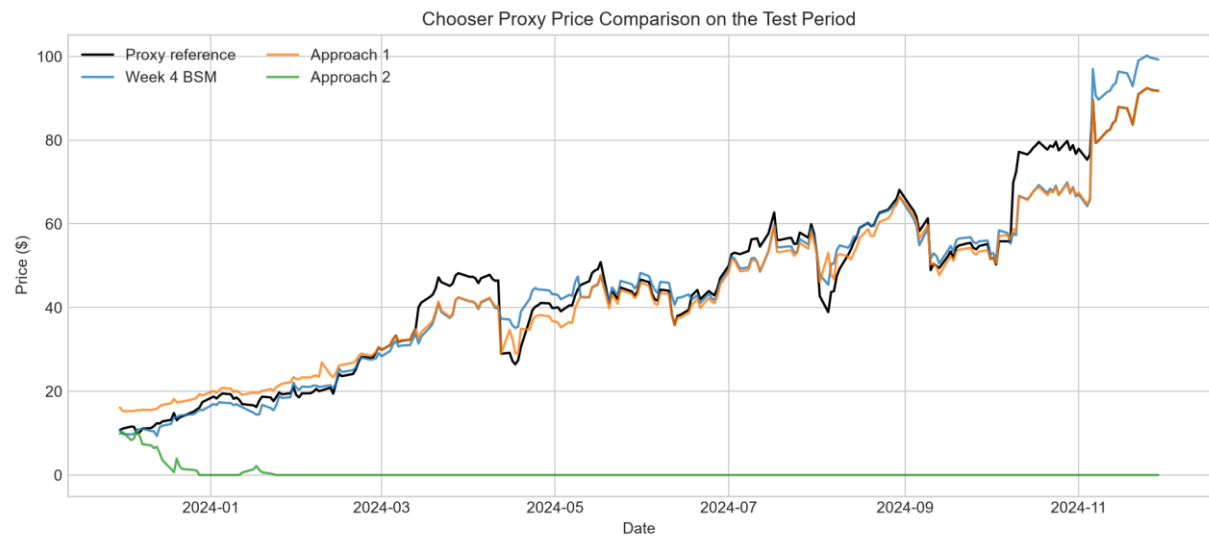
The Week 6 comparison selects Random Forest volatility forecasting followed by BSM. It reduced test RMSE by 11.30% relative to the Week 4 BSM baseline while preserving the same 252 test dates and proxy target.

Selection Method

Candidate hyperparameters were compared using expanding-window cross-validation with a 20-day gap. The separate validation block selected each route. The test period from 2023-11-30 through 2024-11-29 was evaluated once afterward.

Approach	MAE	RMSE	R ²	Mean error
RF vol plus BSM	3.1384	4.3203	0.9575	-1.2781
Trailing 20D BSM	3.5698	4.8706	0.9460	-0.5299
Direct linear price	45.0275	49.9891	-4.6915	-45.0226

Figure 1 Held-out test price comparison



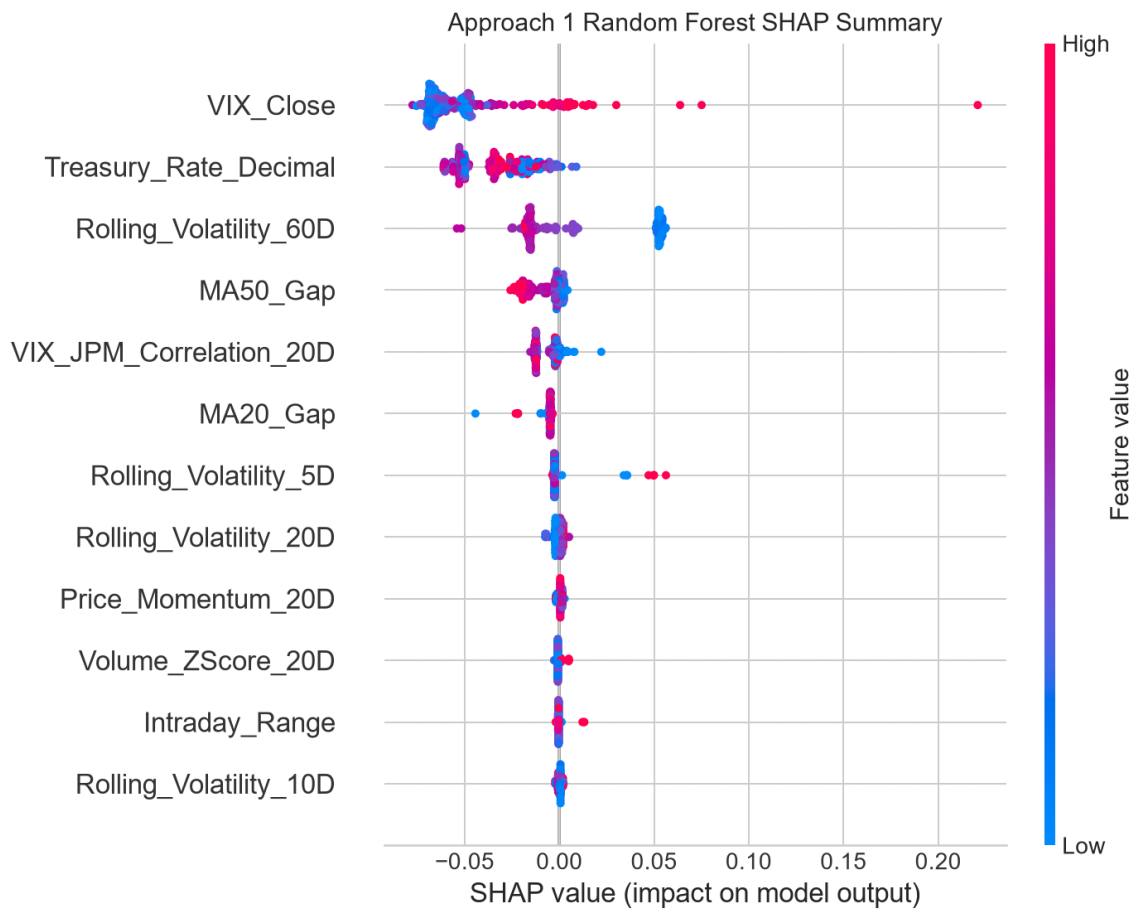
Volatility Forecast

Forecast	MAE	RMSE
Persistence	0.085584	0.120650
Selected Random Forest	0.072424	0.094069

Model Interpretation

The SHAP analysis explains the selected Random Forest in Approach 1 and selected Linear Regression in Approach 2. VIX-related features accounted for 40.80% and 5.06% of total mean absolute SHAP contribution, respectively. These rankings describe fitted behavior and do not identify causal effects.

Figure 2 Approach 1 SHAP summary



Serialized Model Deliverables

Both selected estimators are delivered as native Python pickle files, as required by the Week 6 rubric. Each pickle was reloaded and checked against all 252 held-out rows; the maximum absolute difference from the saved Week 6 inference outputs was below $1e-12$. Joblib copies remain available for the Week 8 application.

Artifact	Selected model	Delivery role
best_volatility_model.pkl	Random Forest volatility forecaster	Required pickle deliverable for Approach 1
best_direct_pricing_model.pkl	Linear Regression direct pricer	Required pickle deliverable for Approach 2
Two .joblib copies	Same fitted estimators	Runtime compatibility for the Week 8 tool

Decision and Limits

- Use Random Forest volatility plus BSM as the selected ML route in Week 8.
- Retain direct Linear Regression as a documented extrapolation failure.
- Display the BSM baseline beside every ML estimate.
- Warn when current inputs exceed the fitted-sample range.
- Describe every score as proxy-target performance.